Week 5 reading guide and worked examples

Weekly overview

Required learning outcomes

By the end of this week, you should be able to:

- use total and implicit differentiation under stated regularity conditions, identify homogeneous functions and apply Euler's theorem (M7);
- classify quadratic forms by definiteness, determine whether sets are convex, and distinguish convexity, concavity and quasiconcavity using valid definitions or derivative tests (M8); and
- derive marginal and conditional distributions from a joint distribution, assess independence, and calculate conditional expectations using iterated expectation or total variance where appropriate (S5).

Prerequisites and retrieval

Retrieve basis and eigenvalue calculations (M5), column gradients and Hessians (M6), conditional probability (S2), and expectation and variance under linear transformations (S4). Questions 9 and 10 contain three explicit closed-note retrieval parts spanning M5, S2 and S4; the current-week calculus tasks also require fresh use of M6.

Before starting, check that you can multiply a matrix by a column vector, compute a 2×2 determinant, differentiate a two-variable function, normalise a simple PMF or PDF, and calculate a weighted expectation.

Student materials

Before the conditional-expectation work, use the required conditional expectation bridge note. It is point-of-need support within the existing reading and worked-example allocation, not additional workload.

1. Follow the exact written locators and exclusions in the reading guide.
2. Study the original worked examples, covering each solution and reproducing its method from a blank page.
3. Attempt every part of the exercise sheet before opening full solutions.
4. Consult the progressive hints only after a serious attempt and stop once you have the next useful step.
5. Mark your work with the complete solutions, recording the first cause of each error by outcome ID.
6. Finish with the 25-minute closed-note self-test, mark it, and schedule a blank-page reattempt of missed parts after two or three days.

The required external material is written and linked at exact section level. No video, applet, code, specialist software or external exercise is required.

Suggested study sequence

1. Write the scalar and system implicit-function conditions at the top of your notes.
2. Study total differentiation and homogeneity, checking the orientation of every gradient and the dimensions of every Jacobian.
3. Work through quadratic forms before moving to convex sets and curvature. Keep concavity and quasiconcavity as distinct classifications.
4. Begin the statistics block by drawing the support of each joint distribution in words or inequalities. Derive marginals before conditionals.
5. Reproduce the conditional-expectation and total-variance calculations, then attempt the exercise sheet in alternating mathematics and statistics blocks.
6. Complete Questions 9 and 10 without earlier notes. Mark all work, update the error log, and use the self-test as the final closed-note check.

Workload and required outputs

The times below are provisional planning guidance, not a guarantee, and have not yet been verified by students. They are subject to first-cohort calibration.

Component	Target time	Required output
Exact required reading and institutional worked-example reproduction	5 h	Original notes plus six reproduced examples, with conditions and supports stated
Week 5 problem sheet	4 h	All 17 scorable parts attempted and marked
Closed-note self-test	25 min	Four 5-mark parts attempted and marked
Immediate error-log update	5 min	Errors labelled `M7`, `M8`, `S5` or the relevant retrieval outcome
Core total	**9 h 30 min**	Completed Week 5 study cycle

The problem sheet contains 10 numbered questions, 17 scorable parts, 100 marks and a 240-minute target. Across marks, it contains 30% foundation, 60% standard and 10% stretch work. Parts whose primary demand is explanation, derivation or synthesis carry 46% of the marks.

End-of-week check

Before moving on, make sure that you can state why an implicit derivative exists locally, use the symmetric part of a matrix in a quadratic form, separate concavity from quasiconcavity, state the full support of a conditional distribution, and distinguish the two components in the law of total variance.

Required reading route

How to use this route

Complete the three required written selections in order. Open only the exact locators below and follow every skip instruction. The 300-minute allocation includes both the linked reading and reproduction of the six original institutional examples in worked_examples.md, as required by the approved Week 5 schedule.

The linked sources remain on their authors' sites. Osborne and Pishro-Nik material is link-only: do not copy its explanations, questions or solutions into your notes. Close each source and write an original summary instead.

If equation rendering or assistive-mathematics access prevents you from using a linked page, skip that page and use this course's original worked examples, hints and solutions for the same outcome. Report the access problem to the course team. To report a broken, inaccessible or unavailable required source, email cristiano.cantore@uniroma1.it with the week, item or source locator. You may request this confidentially without disclosing disability or other sensitive personal information. The programme Coordinator will provide an accessible replacement PDF or an approved alternative reading. You will not be assessed on source-specific wording.

Required bridge note

Before the conditional-expectation selection, read the conditional expectation bridge note. It supplies the course's conditional-expectation conventions within the existing Week 5 reading and worked-example allocation, not additional workload.

1. Total and implicit differentiation and homogeneity (`M7`)

Resource `MATH-M7-OSB-REQ`; source `OSBORNE_MM`. In Martin J. Osborne, *Mathematical Methods for Economic Theory*, read:

- Section 2.3, "Derivatives of functions defined implicitly";
- Section 2.4, "Differentials and comparative statics"; and
- Section 2.5, "Homogeneous functions".

Skip: Section 2.3 Exercises 10–12 and Section 2.5 Exercises 14–15. Do not copy or adapt any Osborne exercise. Do not use an implicit derivative without stating differentiability and the required non-zero derivative or nonsingular Jacobian.

Translate the source to the course notation. Gradients are columns. If $\mathbf{F} : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ defines $\mathbf{x}(\theta)$ locally, then $\mathbf{J}_{\mathbf{F},\mathbf{x}}$ is $n \times n$ and must be nonsingular before it is inverted.

Target for the linked reading and reproduction of Worked Examples 1–2: 75 minutes.

Reading output. State the scalar and system implicit-derivative formulae with their conditions. Give one definition of homogeneity and state Euler's theorem, including differentiability and the domain condition needed for scaling.

2. Quadratic forms and curvature (`M8`)

Resource `MATH-M8-OSB-REQ`; source `OSBORNE_MM`. In Osborne, *Mathematical Methods for Economic Theory*, read:

- Section 3.1 on one-variable curvature;
- Sections 3.2.1, 3.2.2 and 3.2.3 on quadratic forms;
- Section 3.3 on multivariable curvature; and
- Section 3.4 on quasiconcavity.

Skip: every rotatable or interactive figure and proof extension beyond applying the stated operational tests. Do not reproduce or adapt the source's questions. Keep only the set-convexity, quadratic-form, curvature and quasiconcavity material required by `M8`.

When a quadratic form is written with a non-symmetric matrix, replace the matrix by its symmetric part before applying a definiteness test. A global Hessian conclusion requires a convex domain and the stated differentiability conditions. Do not infer concavity merely from quasiconcavity.

Target for the linked reading and reproduction of Worked Examples 3–4: 105 minutes.

Reading output. Write a classification table for positive definite, positive semidefinite, negative definite, negative semidefinite and indefinite quadratic forms. Then give separate definitions of a convex set, a concave function and a quasiconcave function.

3. Joint and conditional distributions (`S5`)

Resource `STAT-S5-REQ-PISHRO`; source `PISHRO_NIK`. In Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes*, read the following non-contiguous sections in order:

- §5.1.1, "Joint Probability Mass Function";
- §5.1.2, "Joint Cumulative Distribution Function";
- §5.1.3, "Conditioning and Independence";
- §5.1.5, "Conditional Expectation";
- §5.2.1, "Joint Probability Density Function";
- §5.2.2, "Joint Cumulative Distribution Function"; and
- §5.2.3, "Conditioning and Independence".

Skip: §5.1.4; every unassigned part of §5.3 on dependence measures or transformations; §5.4 end-of-chapter questions; MATLAB; and all linked interactive material. Approval is limited to this exact external-link route: do not copy, excerpt, adapt, translate, reformat or embed the source's prose, examples, questions or solutions.

For every PMF or PDF, write the joint support before summing or integrating. For a conditional distribution, state both the conditional support and the condition that the marginal denominator is positive.

Target for the linked reading and reproduction of Worked Examples 5–6: 120 minutes.

Reading output. Derive one marginal from a joint PMF and one from a joint PDF. State the factorisation test for independence, the law of iterated expectation and the law of total variance.

Completion check

Your notes should now contain:

- a dimensionally consistent system implicit-derivative formula;
- one verified application of Euler's theorem;
- a definiteness classification with a boundary case;
- one convex-set argument and one quasiconcavity argument;
- a discrete and a continuous marginal/conditional derivation; and
- separate calculations for $\mathbb{E}[\mathrm{Var}(X \mid Y)]$ and $\mathrm{Var}(\mathbb{E}[X \mid Y])$.

Worked examples

These six original institutional examples support `M7`, `M8` and `S5`. Cover each **Solution**, follow the **Method**, and reproduce the derivation from a blank page. The 95 minutes suggested below are included in the five-hour reading-and-example allocation.

Example 1 — total and scalar implicit differentiation (`M7`; 15 minutes)

Let

$$C(a,b) = a^2 b + \log b, \qquad a \in \mathbb{R}, \quad b > 0.$$

Find the total differential at $(a,b) = (1,2)$ and use it to approximate the change when $\mathrm{d}a = 0.1$ and $\mathrm{d}b = -0.05$. Then suppose $F(x,\theta) = x^2 + \theta x - 3 = 0$. Find $\mathrm{d}x/\mathrm{d}\theta$ at $(x,\theta) = (1,2)$.

Method. Form the column gradient before taking its transpose in the total differential. For the implicit derivative, check differentiability and the non-zero partial derivative F_x.

Solution. The column gradient is

$$\nabla C(a,b) = \begin{bmatrix} 2ab \\ a^2 + 1/b \end{bmatrix}.$$

At $(1,2)$,

$$\nabla C(1,2) = \begin{bmatrix} 4 \\ 3/2 \end{bmatrix}.$$

Therefore,

$$\mathrm{d}C = \nabla C(1,2)^{\mathsf{T}} \begin{bmatrix} \mathrm{d}a \\ \mathrm{d}b \end{bmatrix} = 4\,\mathrm{d}a + \frac{3}{2}\,\mathrm{d}b.$$

For the stated changes,

$$\Delta C \approx \mathrm{d}C = 4(0.1) + \frac{3}{2}(-0.05) = 0.325.$$

The polynomial F is continuously differentiable. At $(1,2)$,

$$F_x = 2x + \theta = 4 \neq 0, \qquad F_\theta = x = 1.$$

The scalar implicit-function formula is valid locally, and

$$\left.\frac{\mathrm{d}x}{\mathrm{d}\theta}\right|_{(1,2)} = -\frac{F_\theta}{F_x} = -\frac{1}{4}.$$

Interpretation. A total differential is a first-order approximation, not an exact finite change. The implicit derivative is a local conclusion whose validity depends on the non-zero denominator.

Example 2 — a system derivative and Euler's theorem (M7; 20 minutes)

The equations

$$F_1(u, v, \theta) = u + 2v - \theta - 3 = 0,$$
$$F_2(u, v, \theta) = u^2 + v - 2 = 0$$

hold at $(u, v, \theta) = (1, 1, 0)$. Find $\mathrm{d}[u, v]^\mathsf{T}/\mathrm{d}\theta$ there. Then verify Euler's theorem for $h(x, y) = x^2/y$ on $x > 0, y > 0$.

Method. For the system, identify the 2×2 Jacobian with respect to the two endogenous variables and check that it is nonsingular. For homogeneity, scale both arguments by the same positive number before applying Euler's theorem.

Solution. The relevant derivatives are

$$\mathbf{J}_{\mathbf{F},(u,v)} = \begin{bmatrix} 1 & 2 \\ 2u & 1 \end{bmatrix}, \qquad \mathbf{F}_\theta = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

At the stated point,

$$\det \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = -3 \neq 0.$$

Hence the implicit function exists locally and

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} u'(\theta) \\ v'(\theta) \end{bmatrix} = - \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Solving gives

$$u'(0) = -\frac{1}{3}, \qquad v'(0) = \frac{2}{3}.$$

For $t > 0$,

$$h(tx, ty) = \frac{(tx)^2}{ty} = t\, h(x, y),$$

so h is homogeneous of degree one. It is differentiable on the stated domain, with

$$h_x = \frac{2x}{y}, \qquad h_y = -\frac{x^2}{y^2}.$$

Euler's theorem gives

$$x h_x + y h_y = \frac{2x^2}{y} - \frac{x^2}{y} = \frac{x^2}{y} = h(x, y).$$

Interpretation. Nonsingularity is not a cosmetic matrix check: it is what allows the endogenous-variable derivative to be solved uniquely. Euler's identity expresses the degree of a differentiable homogeneous function.

Example 3 — the symmetric part of a quadratic form (M8; 15 minutes)

Classify

$$q(\mathbf{x}) = \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}, \qquad \mathbf{A} = \begin{bmatrix} 1 & 4 \\ -2 & 3 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

Method. A quadratic form depends only on the symmetric part $(\mathbf{A} + \mathbf{A}^{\mathsf{T}})/2$. Apply a definiteness test to that matrix.

Solution. The symmetric part is

$$\mathbf{S} = \frac{\mathbf{A} + \mathbf{A}^{\mathsf{T}}}{2} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}.$$

Its leading principal minors are

$$1 > 0, \qquad \det(\mathbf{S}) = 3 - 1 = 2 > 0.$$

Thus $\mathbf{S}$ is positive definite. Equivalently,

$$q(\mathbf{x}) = x^2 + 2xy + 3y^2 = (x + y)^2 + 2y^2 > 0$$

for every non-zero $\mathbf{x}$. The quadratic form is positive definite.

Interpretation. Applying a symmetry-based test directly to the original non-symmetric matrix would be invalid. The skew-symmetric part contributes zero to $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}$.

Example 4 — convexity and quasiconcavity are different (M8; 15 minutes)

Show that

$$E = \{(x, y) \in \mathbb{R}^2 : x \geq 0,\ y \geq 0,\ x + 2y \leq 4\}$$

is convex. On the domain $\mathbb{R}^2_{++} := \{(x, y) : x > 0, y > 0\}$, classify $g(x, y) = x/y$ as convex, concave or quasiconcave.

Method. Use the convex-combination definition for the set. Use the Hessian to assess convexity and concavity of the function, but use upper contour sets for quasiconcavity.

Solution. Take $(x_1, y_1), (x_2, y_2) \in E$ and $\lambda \in [0, 1]$. Both coordinates of their convex combination are non-negative. Moreover,

$$\begin{aligned} &\lambda x_1 + (1 - \lambda)x_2 + 2(\lambda y_1 + (1 - \lambda)y_2) \\ &\quad = \lambda(x_1 + 2y_1) + (1 - \lambda)(x_2 + 2y_2) \leq 4\lambda + 4(1 - \lambda) = 4. \end{aligned}$$

Thus the convex combination also lies in E, so E is convex.

For g,

$$\mathbf{H}_g(x, y) = \begin{bmatrix} 0 & -1/y^2 \\ -1/y^2 & 2x/y^3 \end{bmatrix},$$

whose determinant is

$$-\frac{1}{y^4} < 0.$$

The Hessian is indefinite throughout the domain, so g is neither convex nor concave.

For $c > 0$, the upper contour set is

$$\{(x, y) \in \mathbb{R}^2_{++} : g(x, y) \geq c\} = \{(x, y) \in \mathbb{R}^2_{++} : x - cy \geq 0\}.$$

This is the intersection of the convex domain with a half-space, so it is convex. For $c \leq 0$, the upper contour set is the whole domain, which is convex. Therefore g is quasiconcave.

Interpretation. Quasiconcavity concerns convex upper contour sets. It does not require a negative semidefinite Hessian and does not imply concavity.

Example 5 — a discrete joint distribution (S5; 15 minutes)

The joint PMF of $X \in \{0, 1, 2\}$ and $Y \in \{0, 1\}$ is

x	$p_{X,Y}(x, 0)$	$p_{X,Y}(x, 1)$
0	1/4	0
1	1/4	1/4
2	0	1/4

Find $\mathbb{E}[X \mid Y]$ and verify the law of iterated expectation.

Method. Sum the joint PMF to obtain the marginal of Y, divide each column by its positive marginal, and calculate the conditional means.

Solution. The marginal probabilities are

$$\mathbb{P}(Y = 0) = \frac{1}{2}, \qquad \mathbb{P}(Y = 1) = \frac{1}{2}.$$

Conditional on $Y = 0$, the probabilities of $X = 0$ and $X = 1$ are each $1/2$, so

$$\mathbb{E}[X \mid Y = 0] = \frac{1}{2}.$$

Conditional on $Y = 1$, the probabilities of $X = 1$ and $X = 2$ are each $1/2$, so

$$\mathbb{E}[X \mid Y = 1] = \frac{3}{2}.$$

Thus

$$\mathbb{E}[X \mid Y] = \begin{cases} 1/2, & Y = 0, \\ 3/2, & Y = 1. \end{cases}$$

Iterating gives

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \frac{1}{2}\frac{1}{2} + \frac{3}{2}\frac{1}{2} = 1.$$

The marginal PMF of X is

$$p_X(0) = \frac{1}{4}, \qquad p_X(1) = \frac{1}{2}, \qquad p_X(2) = \frac{1}{4},$$

so $\mathbb{E}[X] = 0(1/4) + 1(1/2) + 2(1/4) = 1$, confirming the identity.

Interpretation. The conditional expectation is a random variable because its value depends on the realised value of Y.

Example 6 — continuous conditioning and total variance (`S5`; 15 minutes)

Let the joint density be

$$f_{X,Y}(x, y) = 1 \quad \text{for } 0 < x < 1, \quad x < y < x + 1,$$

and zero otherwise. Derive $f_{Y|X}(y \mid x)$, $\mathbb{E}[Y \mid X = x]$, and $\mathrm{Var}(Y)$ by total variance.

Method. Check the joint density over its sloping unit-width support. Conditional on $X = x$, retain the shifted support $x < y < x + 1$. Then separate the two terms in the law of total variance.

Solution. The proposed density is valid because it is non-negative and

$$\int_0^1 \int_x^{x+1} 1 \, \mathrm{d}y \, \mathrm{d}x = 1.$$

For $0 < x < 1$,

$$f_X(x) = \int_x^{x+1} 1 \, \mathrm{d}y = 1.$$

The marginal density is zero otherwise. Therefore,

$$f_{Y|X}(y \mid x) = \frac{1}{1} = 1, \qquad x < y < x + 1,$$

for $0 < x < 1$, and the conditional density is zero outside that conditional support. Hence $Y \mid X = x$ is uniform on $(x, x + 1)$, so

$$\mathbb{E}[Y \mid X = x] = x + \frac{1}{2}, \qquad \mathrm{Var}(Y \mid X = x) = \frac{1}{12}.$$

Because $X \sim \mathrm{Uniform}(0, 1)$,

$$\mathrm{Var}(X) = \frac{1}{12}.$$

The first component of total variance is

$$\mathbb{E}[\mathrm{Var}(Y \mid X)] = \frac{1}{12},$$

and the second is

$$\mathrm{Var}(\mathbb{E}[Y \mid X]) = \mathrm{Var}(X + 1/2) = \mathrm{Var}(X) = \frac{1}{12}.$$

Therefore,

$$\mathrm{Var}(Y) = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}.$$

Interpretation. Total variance adds expected variation within conditional groups to variation between their conditional means. The support must be carried through every density and conditional calculation.

Reproduction check

Before starting the exercise sheet, reproduce the nonsingularity check, Euler identity, symmetric-part calculation, upper-contour argument, iterated expectation and both total-variance terms. If any condition or support is missing, repair it before looking at the exercise solutions.